

Beyond the Leaderboard: Informative Missingness and Thin-File Borrowers in Home Credit Default Prediction

ZHANG Qi (21358199) DU Yunhe (21419400)
TANG Yunshu (21275921) WEI Fanjing (21354557)

Kaggle team: msdm5054-zhang-du-tang-wei
MSDM5054 Statistical Machine Learning, Project 1

Abstract

We study default prediction on the Home Credit Default Risk data (307,511 loans, default rate 8.07%), asking not only how accurately default can be predicted but for whom and why. A gradient-boosted model on the application table and six auxiliary tables reaches a 5-fold out-of-fold (OOF) AUC of 0.7926 (Kaggle private score 0.79295), compared with 0.7505 for a regularized logistic regression. We then pose two scientific questions. First, is missingness informative? A model that sees *only which fields are missing* attains AUC 0.5849, well above chance, yet making that information explicit changes predictive performance by less than 0.002 AUC — the missingness signal is almost entirely redundant given the observed covariates. Second, does the model serve “thin-file” applicants, the 14.3% with no credit-bureau history, as well as others? They default more often (10.1% versus 7.7%) and are ranked less accurately (AUC 0.779 versus 0.794), but alternative data helps them substantially more (AUC gain +0.032 versus +0.023, non-overlapping bootstrap intervals): it narrows the gap without closing it. We close by discussing selection bias in the observed loan population, which limits what any model trained on these data can claim.

1 Introduction

Credit scoring for applicants with limited credit history is both a statistical and a social problem. Home Credit, a lender focused on the underbanked, supplements application data with alternative sources such as previous applications, point-of-sale and installment payment histories. Most public solutions to the associated Kaggle competition focus on maximizing AUC through feature engineering and ensembling. We treat predictive performance as the starting point and ask three questions:

- **RQ1 (Missingness).** Many application fields are missing for large fractions of applicants. Is the fact that a field is missing itself predictive of default, and how should missing values therefore be handled?
- **RQ2 (Thin-file applicants).** 14.3% of applicants have no record in the credit-bureau data. Are discrimination and calibration worse for them, and does alternative data narrow the gap?
- **RQ3 (Drivers).** Which features drive predictions, and are their effects directionally plausible?

Our contributions are (i) a reproducible, leakage-free pipeline aggregating all seven tables; (ii) a systematic comparison of missing-data strategies; (iii) a subgroup analysis of discrimination and calibration with bootstrap intervals; and (iv) a discussion of how sample selection limits the conclusions.

2 Data

The training set contains 307,511 applications with a binary default label; the test set contains 48,744 unlabeled applications. Six auxiliary tables link to applications by client ID: bureau records of

credits at other institutions and their monthly balances, previous Home Credit applications, and monthly histories of point-of-sale/cash loans, installment payments and credit cards. Two anomalies are handled explicitly: `DAYS_EMPLOYED` equals 365,243 for about 18% of applicants (a placeholder, predominantly for pensioners), which we set to missing while keeping a binary flag; and `CODE_GENDER` has a few XNA values, set to missing. Missingness is pervasive: 61 of the 131 application-level features are missing for between 1% and 99% of applicants, 51 of them for more than 20% and 42 for more than half, with the building-description fields (e.g. `COMMONAREA_AVG`, 70%) and `EXT_SOURCE_1` (56%) the most affected.

3 Methods

Features. From the application table we build domain ratios (credit/income, annuity/income, annuity/credit, goods price/credit, employment length/age, income per family member), summaries of the three external scores, and the count of missing fields. Each auxiliary table is aggregated to the client level (mean, max, min or sum of numeric fields; record counts; shares of categorical levels; and, for installments, days past due, days before due, payment shortfall and share of late payments). The design has 339 features (131 from the application table).

Models. (a) L_2 -regularized logistic regression ($C = 0.05$) with median imputation plus missingness indicators, one-hot encoding and standardization; (b) LightGBM [Ke et al., 2017] with learning rate 0.03, 32 leaves, at least 100 samples per leaf, row subsampling 0.8, column subsampling 0.5, L_2 penalty 5 and early stopping (200 rounds). All models use 5-fold stratified cross-validation; we report OOF AUC and average fold models for test predictions.

RQ1. For every field with 1–99% missing we compare the default rate when missing versus observed. We fit a logistic model on binary missingness indicators alone, and compare four strategies on application features: median imputation and median imputation plus indicators (logistic), and LightGBM with native missing-value handling versus after median imputation.

RQ2. Thin-file applicants are those with no bureau record. Per group we compute OOF AUC and Brier score for the application-only and all-tables LightGBM models, the AUC gain from auxiliary tables with a 95% bootstrap interval (300 resamples), and decile calibration curves.

RQ3. Exact TreeSHAP contributions [Lundberg et al., 2020] via LightGBM’s `pred_contrib` on a fixed sample of 50,000 applications, ranked by mean absolute contribution.

4 Results

Table 1: Predictive performance (5-fold OOF AUC) and Kaggle scores. The competition closed in 2018, so our entry is a late submission (team `msdm5054-zhang-du-tang-wei`): scored, but not shown on the leaderboard.

Model	Features	OOF AUC	Kaggle public	Kaggle private
Logistic regression (L2)	application	0.7505	–	–
LightGBM	application	0.7686	–	–
LightGBM	all tables	0.7926	0.79713	0.79295

Performance. Two increments are visible in Table 1. Replacing logistic regression by gradient boosting on the same application features adds 0.018 AUC ($0.7505 \rightarrow 0.7686$), indicating that non-linearities and interactions among application fields carry real signal that a linear log-odds model cannot express. Adding the six auxiliary tables adds a further 0.024 AUC ($0.7686 \rightarrow 0.7926$), a larger gain than the change of model class: what the applicant did in the past with other lenders and with Home Credit matters more than a richer functional form on the application form alone.

RQ1: missingness is informative but redundant. Missingness is clearly associated with the outcome: a logistic model using *only* 61 binary missingness indicators, with no observed values at all, reaches AUC 0.5849 (Table 2); Figure 1 shows the fields whose presence or absence separates defaulters most strongly.

Table 2: RQ1: missing-data strategies on application features (OOF AUC).

Strategy (application features)	OOF AUC
Missingness indicators only (logistic)	0.5849
Logistic, median imputation	0.7493
Logistic, median imputation + indicators	0.7505
LightGBM, native NaN handling	0.7686
LightGBM, median-imputed	0.7684

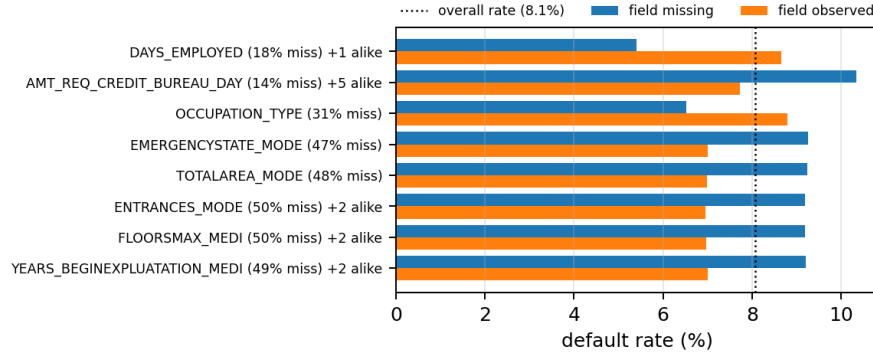


Figure 1: RQ1: default rate when a field is missing versus observed, for the fields with the largest difference.

How it is handled, however, barely matters. Adding missingness indicators to the logistic model moves AUC from 0.7493 to 0.7505 (+0.0012), and LightGBM with native missing-value handling (0.7686) is indistinguishable from LightGBM trained after median imputation (0.7684; the difference, -0.0002 , is far smaller than the spread across folds). This is a redundancy result rather than a negative one: missingness carries information about default, but essentially all of it is recoverable from the observed covariates, so an explicit encoding adds nothing — a caution against adding indicators reflexively and reporting the result as evidence that “missingness is informative”. The observed data cannot distinguish missing-at-random from missing-not-at-random, because the mechanism depends on unobserved values [Little and Rubin, 2019].

Table 3: RQ2: discrimination and calibration by group (OOF). Gain = AUC(all tables) — AUC(application only).

Group	<i>n</i>	Default	AUC (app)	AUC (all)	Gain [95% CI]	Brier (all)
all	307,511	8.07%	0.7686	0.7926	+0.0239 [+0.0222, +0.0256]	0.0655
thin-file (no bureau)	44,020	10.12%	0.7475	0.7794	+0.0319 [+0.0275, +0.0364]	0.0803
with bureau history	263,491	7.73%	0.7708	0.7936	+0.0227 [+0.0211, +0.0242]	0.0630

RQ2: thin-file applicants. Thin-file applicants are 14.3% of the sample and default markedly more often (10.1% versus 7.7%). The model ranks them less accurately: with the full feature set, AUC is 0.7794 for thin-file applicants against 0.7936 for those with bureau history. The auxiliary tables, however, help the thin-file group more — a gain of +0.0319 [0.0275, 0.0364] against +0.0227 [0.0211, 0.0242] for the rest, with non-overlapping bootstrap intervals. Alternative data therefore does what it is advertised to do, narrowing the performance gap for applicants without traditional credit records, but it does not close it: even after adding every auxiliary table, discrimination for thin-file applicants remains below the application-only performance achieved for applicants with bureau history. Brier scores are not comparable across groups here because the base rates differ, so calibration is read from Figure 2(a) instead.

RQ3: drivers. The mean of the three external credit scores dominates by a wide margin (mean [contribution] 0.32 on the log-odds scale, three times the next feature), followed by demographic and loan-structure variables (CODE_GENDER, AMT_ANNUITY, PAYMENT_RATE, GOODS_CREDIT) and

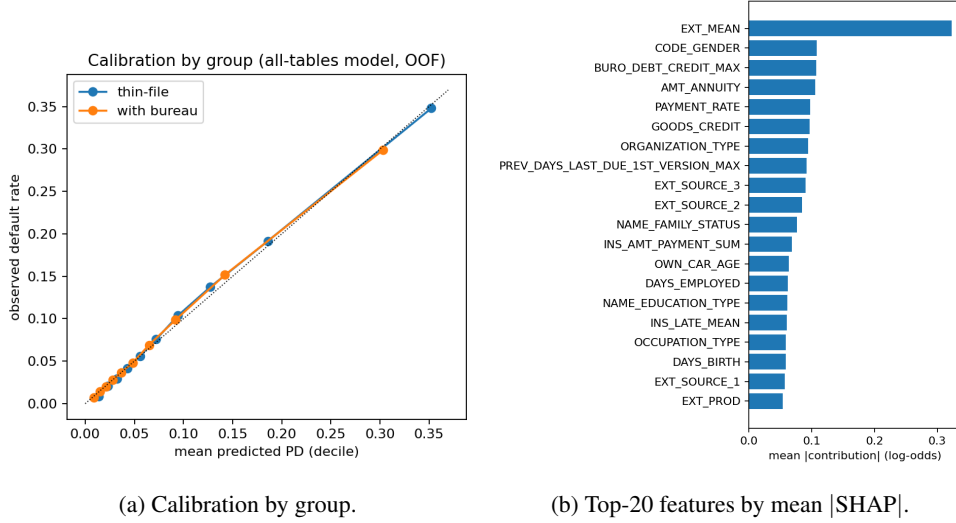


Figure 2: RQ2 calibration and RQ3 feature contributions.

by aggregates of the auxiliary tables (`BURO_DEBT_CREDIT_MAX`, the timing of previous instalment schedules, total amounts paid, and the share of late instalments). The directions are economically sensible: higher external scores and lower debt-to-credit ratios lower predicted risk, while a higher annuity-to-credit ratio — a shorter, heavier repayment schedule — raises it. Two observations connect back to RQ1. First, the individual external scores still appear below `EXT_MEAN`, i.e. the model mostly uses their average, which is defined even when one of them is missing. Second, no missingness-derived feature (`N_MISSING_APP`, `DAYS_EMPLOYED_ANOM`) enters the top twenty, which is exactly what the redundancy result in the RQ1 results above predicts.

SHAP profiles: thin-file vs. with bureau. Repeating the RQ3 decomposition separately for the two RQ2 groups (Figure 3) shows *where* the thin-file discrimination deficit comes from. For applicants with bureau history, the single most relied-upon bureau feature, `BURO_DEBT_CREDIT_MAX` (mean $|SHAP|$ 0.118), is close to unusable for thin-file applicants (0.044, a gap of -0.074), and reliance on `EXT_SOURCE_3` drops by a similar margin ($0.098 \rightarrow 0.041$), largely because that score is itself missing for a large share of thin-file applicants. The model compensates by leaning harder on what it can still observe: the averaged external score `EXT_MEAN` ($0.314 \rightarrow 0.375$, $+0.061$), `EXT_SOURCE_2` ($+0.018$), and — notably — the Home-credit-internal alternative-data aggregates `INS_AMT_PAYMENT_SUM` and `INS_LATE_MEAN` ($+0.014$ each), which thin-file applicants *can* have because the thin-file definition only excludes Credit-Bureau records, not previous Home Credit activity. This pattern is consistent with the RQ2 gain for thin-file applicants being driven mainly by internal behavioural data rather than by bureau data — mean $|SHAP|$ measures reliance, not a decomposition of the AUC gain, so we read it as corroboration rather than proof. It sharpens the RQ2 interpretation: what helps financially excluded applicants is not more credit-bureau information (which does not exist for them) but alternative in-house payment histories.

The dependence profiles (Figure 4) confirm that the dominant effects are monotone and economically sensible, with two caveats worth flagging. `EXT_MEAN` ($\rho = -0.99$) is almost perfectly linear in the log-odds — the model treats the external score average as a near-deterministic risk scale. `BURO_DEBT_CREDIT_MAX` ($\rho = +0.73$) is flat and mildly risk-reducing below a ratio of 1, then jumps sharply once recorded debt exceeds the credited amount — a plausible distress signal, though the small cluster of values far above 1 (clipped in the figure) most likely reflects reporting glitches rather than genuine behaviour, and the model may be reading noise there. The applicant’s gender enters as an almost binary split (male $+0.1$ – 0.3 , female -0.1 – -0.15 log-odds): it is the second most important feature overall, which is statistically unsurprising but raises a fairness concern for deployment, since gender is a protected attribute in most lending regulation; we report it as a property of the data-generating process rather than endorse its use. Finally, `AMT_ANNUITY` ($\rho = +0.94$) and `GOODS_CREDIT` ($\rho = -0.91$) act as smooth, monotone loan-structure controls; neither shows

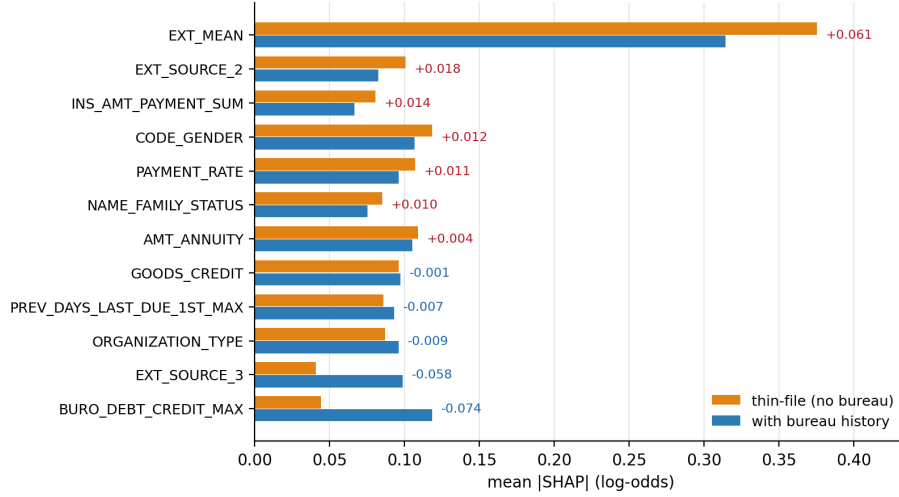


Figure 3: Group-wise TreeSHAP reliance (fold-0 model, 50,000-row sample, of which 7,214 thin-file). Bars are mean |SHAP| within each group; the figure at the right of each pair is the thin-file minus with-bureau difference (red: the model relies on the feature more for thin-file applicants).

a sign reversal, so no interaction pathology is suspected at the top of the ranking.

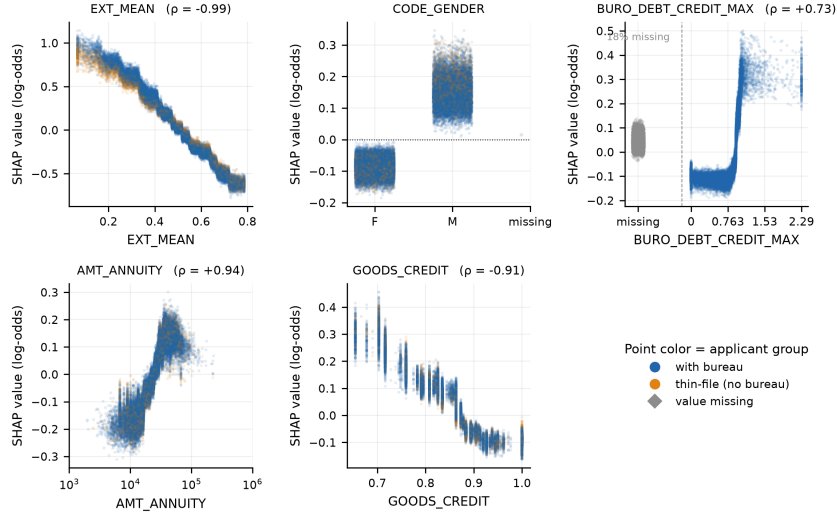


Figure 4: SHAP dependence for the five most important features (colour: applicant group; grey diamonds: feature value missing; continuous features clipped to their 0.5–99.5th percentile for display; ρ = Spearman correlation of feature vs. SHAP among observed values).

Alternative-data source ablation. Is the value of alternative data spread evenly across sources? Starting from the full 339-feature design we removed one auxiliary source at a time, holding feature engineering, hyperparameters, seed and folds fixed (Table 4, Figure 5). Bureau records dominate (-0.0059 AUC when removed), followed by instalment payments (-0.0030) and previous applications (-0.0025); credit cards contribute little (-0.0014) and POS/CASH nothing measurable ($+0.0003$), which we do not read as harmful given the single validation split. The value of alternative data is therefore heterogeneous — and, with the group-wise SHAP profiles, this is also why thin-file applicants lose out: the most valuable source overall is exactly the one they lack.

Robustness. All results above use seed 42. We repeated the two LightGBM configurations and the RQ1 logistic comparison under five seeds (Table 5).¹ The all-tables AUC is stable at $0.7932 \pm$

¹The robustness runs were executed on a LightGBM build that rejects pandas categorical columns, so categorical features

Table 4: Leave-one-source-out ablation of the auxiliary data sources. $\Delta \text{AUC} = \text{AUC}(\text{full}) - \text{AUC}(\text{without source})$.

Removed source	OOF AUC	ΔAUC
None (full model)	0.7926	0.0000
Bureau	0.7867	+0.0059
Installments	0.7896	+0.0030
Previous applications	0.7901	+0.0025
Credit cards	0.7912	+0.0014
POS/CASH	0.7929	−0.0003

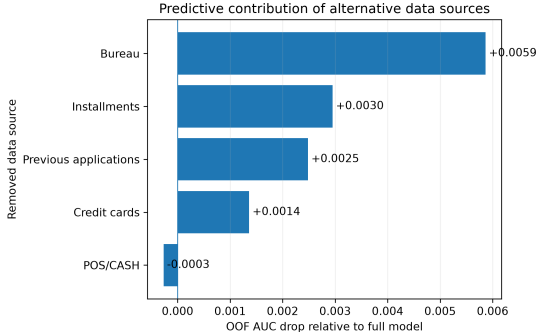


Figure 5: Decrease in OOF ROC-AUC after removing each alternative data source.

0.0003, the thin-file gain exceeds the with-bureau gain in every seed with non-overlapping intervals, and the indicator boost stays within 0.0011–0.0012. Six nearby hyperparameter settings span only 0.0013 AUC (Table 6), so neither conclusion depends on the seed or the parameter choice.

5 Discussion

Selection bias. Labels exist only for loans that were approved. Rejected applicants, possibly including many thin-file applicants, never appear with an outcome. A model trained on these data learns default risk *conditional on past approval*, not in the applicant population; this is the reject-inference problem, a form of sample selection [Heckman, 1979, Crook and Banasik, 2004]. Our RQ2 conclusions therefore concern approved thin-file applicants; performance on currently rejected applicants is not identified from these data.

Missingness mechanism. Association between missingness and default does not identify the mechanism. Distinguishing MAR from MNAR requires additional structure (e.g. shadow variables or instruments), which we leave to future work.

Fairness and inclusion. Two facts from the RQ2 results above interact badly. Thin-file applicants default at 10.1% against 7.7% for the rest, and they are ranked less accurately. A single approval threshold applied to both groups therefore concentrates the model’s errors on the group that is already the hardest to serve: at any fixed cutoff, a larger share of creditworthy thin-file applicants is rejected and a larger share of risky ones accepted than for applicants with bureau history. This reflects having less information about these applicants rather than a defect of the model — the gap alternative data is meant to fill, and on our evidence only partly does. Calibration is not the problem: both groups track the diagonal closely in Figure 2(a), with a mild under-prediction in the middle deciles common to both. The deficit is one of *ranking*, not *level*: the probabilities mean what they say in both groups, but within the thin-file group the model is less able to tell the risky from the safe.

Limitations. Without a large-scale hyperparameter search the absolute AUCs sit slightly below

were passed as integer codes; the seed-42 column therefore differs from Table 1 by at most 0.0014 AUC. All other results, including the ablation, use the native categorical handling of Table 1.

Table 5: Multi-seed robustness. Each seed reseeds both the folds and the model fitting. RQ2 gains are $\text{AUC}(\text{all tables}) - \text{AUC}(\text{application only})$ within a group; the RQ1 boost is the change in logistic AUC from adding missingness indicators.

Metric	Seed=42	Seed=2026	Seed=7	Seed=123	Seed=2024	Mean $\pm$ Std
LGB app-only OOF AUC	0.7698	0.7686	0.7694	0.7690	0.7695	0.7693 ± 0.0005
LGB all-tables OOF AUC	0.7930	0.7931	0.7930	0.7936	0.7935	0.7932 ± 0.0003
RQ2 gain: thin-file	+0.0305	+0.0330	+0.0309	+0.0315	+0.0319	$+0.0316 \pm 0.0010$
RQ2 gain: with bureau	+0.0220	+0.0232	+0.0225	+0.0235	+0.0228	$+0.0228 \pm 0.0006$
RQ1: logit +indicator boost	+0.0012	+0.0011	+0.0012	+0.0012	+0.0012	$+0.0012 \pm 0.0000$

Table 6: Hyperparameter neighbourhood check (3-fold CV, all tables). The first row is the production setting; all six fall within a 0.0013 AUC band.

Setting	num_leaves	min_child	colsample	reg_lambda	OOF AUC
current	32	100	0.5	5.0	0.7918
deeper	63	100	0.5	5.0	0.7910
more conservative	15	200	0.5	5.0	0.7923
colsample 0.8	32	100	0.8	5.0	0.7916
reg_lambda 20	32	100	0.5	20.0	0.7921
kaggle common	31	30	0.7	1.0	0.7911

the best published solutions, though the comparisons are like-for-like and stable across seeds and settings. Auxiliary tables are aggregated over each client’s entire history, discarding recency; and results come from one lender and one period, so the size of the thin-file gap need not transfer.

6 Conclusion

Gradient boosting on all seven tables reaches an out-of-fold AUC of 0.7926 (Kaggle private 0.79295), alternative data contributing more than the model class. Missingness predicts default on its own (AUC 0.5849) yet is redundant given the observed covariates, so its handling moves performance by less than 0.002 AUC. The distributional question matters more: applicants with no bureau history default more often, are ranked less accurately, and gain roughly 40% more from alternative data — narrowing, not closing, the gap. For a lender whose mission is financial inclusion, the leaderboard metric is the wrong summary: an aggregate AUC gain can coexist with persistently worse service for the applicants the product is meant to reach.

Contributions. ZHANG Qi: the research design and three research questions, the seven-table feature pipeline, the RQ1–RQ3 experiments, the Kaggle submission, and the main text. DU Yunhe: the interpretability analysis — group-wise TreeSHAP profiles, dependence diagnostics and the figures (Section 4.4). TANG Yunshu: the robustness and sensitivity analysis — a five-seed replication of the main models and the six-setting hyperparameter neighbourhood check (Tables 5–6). WEI Fanjing: the alternative-data ablation — leave-one-source-out experiments across the five auxiliary sources and the marginal AUC contribution analysis (Table 4, Figure 5). All authors reviewed the manuscript. Code: <https://github.com/ZQ-77777/msdm5054-project1>.

Use of AI tools. An LLM assistant (Claude, Anthropic) helped refine the research questions and draft initial versions of the code and of this report’s outline. All code was reviewed and run by the team; every reported number and figure comes from our own pipeline, and the interpretations are ours.

References

- J. Crook and J. Banasik. Does reject inference improve application scoring models? *Journal of Banking & Finance*, 28(4):857–874, 2004.
- J. J. Heckman. Sample selection bias as a specification error. *Econometrica*, 47(1):153–161, 1979.
- G. Ke et al. LightGBM: A highly efficient gradient boosting decision tree. In *NeurIPS*, 2017.
- R. J. A. Little and D. B. Rubin. *Statistical Analysis with Missing Data*. Wiley, 3rd ed., 2019.
- S. M. Lundberg et al. From local explanations to global understanding with explainable AI for trees. *Nat. Mach. Intell.*, 2:56–67, 2020.